



ABCfold NPF Conformation Pipeline

Six structure-prediction backends, launched together, clustered by TM-helix alignment

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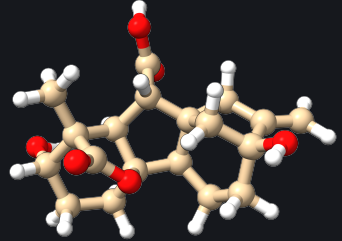
August 7, 2026

Outline

1. Background
2. Motivation
3. Pipeline overview
4. Pre-processing
5. Processing
6. Post-processing
7. Status & next steps

Gibberellins

- ▶ Key phytohormone in plants; isolated in 1931 from *Gibberella fujikuroi* (fungus), later found in plants (1950)
 - ▶ Promotes growth and development: stem elongation, seed germination, flowering
 - ▶ GA-deficiency mutants (key oxidase disrupted): smaller but more resilient, **higher grain yield**
- ⇒ GA-deficiency-linked yield gains underpinned the **Green Revolution** (1970s)



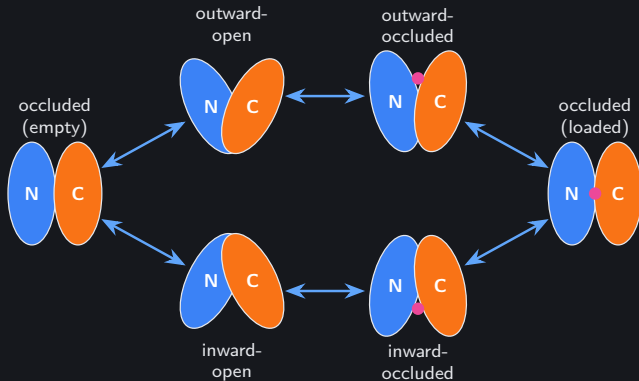
Gibberellin A1 — $\text{C}_{19}\text{H}_{24}\text{O}_6$, 348.4 g/mol
(PubChem CID 5280379)

Nitrate and Peptide Transporters (NPF)

- ▶ Part of the **Major Facilitator Superfamily (MFS)** of membrane transporters
- ▶ Transport nitrate (NO_3^-), dipeptides, and gibberellins
- ▶ Tissue- and substrate-specific

MFS transport cycle: six conformations

Alternating-access mechanism: transport requires passing through a fully **occluded** intermediate in either direction — the **biological reason** the pipeline targets six conformations per protein, not six independent opinions on one.



What's experimentally known

- ▶ Only 2 **NPFs** have experimentally resolved structures: NPF6.3 (4OH3, 5A2N, 5A2O), NPF2.10 (9UI1, 9UI6, 9UIF, 9UIT)
- ▶ Only **importers** are well characterized in the literature (technical constraints)
- ▶ **Exporters** remain unidentified



NPF6.3 — PDB 4OH3, X-ray, 3.25 Å
Sun *et al.*, *Nature* 2014

Research project goals

1. **Ab initio modelling of 3D structures**

1.1 Pipeline to generate the six conformations

1.2 Network interaction analysis (ligand and intra-chain)

(this project)

2. **Functional annotation of NPFs**

2.1 Identify key features of importers vs. non-importers

2.2 Eventually, find exporters

The question

- ▶ Arabidopsis **NPF** (Nitrate/Peptide Transporter) family
Which conformations can each NPF transporter adopt, apo and ligand-bound?
- ▶ Structure prediction as a proxy for conformational sampling

Why not just one model?

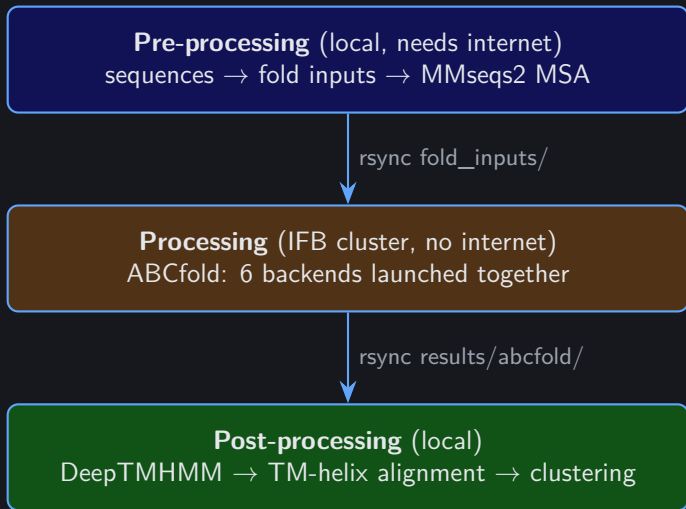
- ▶ Sibling repo AF3_NPF_pipeline already sampled **30 AF3 seeds/protein**
- ▶ Overlaying those ensembles with Boltz-2 (Gibberellin importers, 8 proteins) showed: AF3 alone **does not** recover every conformation a second, architecturally-different model finds
- ⇒ One model isn't enough — run **six together** instead of resampling one

The six backends

- ▶ AlphaFold3
- ▶ Boltz-2
- ▶ Chai-1
- ▶ OpenFold3
- ▶ Protenix
- ▶ RosettaFold3

All launched **together** per protein via ABCfold's `abcfold -abcopr` against the *same* resolved fold input.

Three stages



Stage 1–3: from sequence to resolved fold input

1. **Sequences** — UniProt → per-protein FASTA
2. **Fold input** — `fold_input.json` per protein × apo/holo (AF3-dialect JSON, ABCfold's native input: sequence + every replica seed)
3. **MMseqs2 default run** — MSA + top-hit templates from the ColabFold webserver, no manual curation, no pocket restraint; embedded into `fold_input.resolved.json`

Runs locally, driven by Snakemake (`workflows/preprocessing/Snakefile`); resumable — existing resolved inputs are never re-fetched.

Stage 4: ABCfold on the IFB cluster

- ▶ One `abcfold -abcopr...` call per protein $\times$ form
- ▶ Launches all 6 backends against the same `fold_input.resolved.json`
- ▶ AlphaFold3 backend runs via IFB's `alphafold` module (Singularity wrapper, auto-discovered) — no separate image needed
- ▶ Boltz/Chai-1/OpenFold3/Protenix/RosettaFold3 built once via `--prime` on a login node (needs internet)

Gibberellin importers run first

- ▶ 8 GA1-ligand proteins (NPF2.1, NPF2.5, NPF2.10, NPF2.12, NPF2.13, NPF3.1, NPF4.1, NPF5.6) — the group already studied with AF3 + Boltz-2
- ▶ Listed in `priority_gibberellin.txt`, put first in the SLURM array manifest
- ▶ Fastest path to validating the six-backend approach against a known result

Stage 5–6: topology and alignment

1. **DeepTMHMM** — TM-helix topology (BioLib), once
2. **TM-helix Kabsch alignment** — per protein $\times$ form, pooled across every backend and seed

Output: `results/tm_alignment/<protein>/{aligned_ca_tm.npy, meta.csv}`
— `meta.csv` now carries a `model` column (`alphafold3/boltz/chai1/openfold3/protenix/rosettafold3`).

Stage 7: clustering & visualisation

- ▶ Notebook-based (JupyterLab, envs/notebook.yaml)
- ▶ Same schema as AF3_NPF_pipeline's clustering notebooks + one new axis: **color/facet by backend**
- ▶ *That's the whole point of this pipeline:* which backend(s) actually reach which conformational cluster?
- ▶ Status:

Where we are



Next steps



Questions?